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DIVERSITY AND DISTRIBUTION OF ORTHOPTERA FAUNA IN LOTHIAN ISLAND WILDLIFE SANCTUARY, SUNDERBAN BIOSPHERE RESERVE, WEST BENGAL

D. SURESH CHAND, C. RAGHUNATHAN, RITIKA DATTA, RANITA DATTA, ARYA SEN, AND SWAPANKR. DAS.

Zoological Survey of India, M-Block, New Alipore, Kolkata- 700053, West Bengal, India.
e-mail: dschand_manish@rediffmail.com

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ABSTRACT: Diversity and distribution of the Orthoptera fauna in Lothian Island Wildlife Sanctuary within the Sunderban Biosphere Reserve, West Bengal was conducted in three different zones. One hundred fifty one Orthoptera specimens were collected from various locations across Lothian Island and the study led to the identification of 30 species belonging to 25 genera under 15 sub families and 6 families, viz., Acrididae, Pyrgomorphidae, Tetrigidae, Gryllidae, Trigonidiidae, and Tettigoniidae. Acrididae and Tettigoniidae were relatively more represented. All the 30 species are reported for the first time from Lothian Island, of which five species are new to Sunderban Biosphere Reserve viz., *Conocephalus (Anisoptera) maculatus* (Le Guillou, 1841), *Tagasta indica* Boliver, 1905, *Xenocatantops humilis* (Serville, 1838), *Hebarditettix quadratus* (Hancock, 1915), and *Loxoblemmus equestris* Saussure, 1877. While three species are reported for the first time from the state of West Bengal, *Ducetia dichotoma* Ingrisch and Shishodia, 1998, *Hexacentrus unicolor* Serville, 1831, and *Pusana laevis* (Uvarov, 1921).

Key words: Orthoptera, Biodiversity, Distribution, Lothian Island, Sunderban Biosphere Reserve.

INTRODUCTION

Sunderban is the largest single mangrove belt of the world comprising a total area of 9827 sq.km., which lies in both India and Bangladesh, while the Indian part alone constitutes 4264 sq.km and it lies in between latitudes 21°10'46" N and longitudes 88°58'21"E. Sunderban was declared as National Park, Biosphere Reserve, UNESCO World Heritage site and also withholds three Wildlife Sanctuaries, namely, Sajnekhali Wildlife Sanctuary (WLS), Halliday Island WLS, and Lothian Island WLS. Lothian Island WLS with an area of 38 km² was declared as the Wildlife Sanctuary in the year 1976 and extends between 21°32'50"N to 21°42'30" N latitude and 88°18'10" E to 88°21'30" E longitude. Lothian WLS is one among the islands located in the southern tip of Sunderbans, close to the Bay of Bengal (BRAHMA *et al.*, 2012). Orthopterans are one among the invertebrate groups of insects and so far there have been no studies on this group were made from this sanctuary. However, earlier work by AGARWAL and GHOSE (1995) reported 10 species of Orthoptera, DAS *et al.* (2016) have reported 38 species, under 32 genera, belonging to 15 sub families and 7 families from some areas of Sunderbans viz., Bakkhali, Sagar island, Gosaba, and Bali island, and World Wide Fund for Nature-India reported 45 species belonging to 36 genera and 8 families of Orthoptera in its report on faunal diversity of Sunderban on 2017 (DANDA *et al.*, 2017). The total number of Orthopteran known from India are 1166 species belonging to 449 genera under 22 families (GUPTA and CHANDRA, 2019; TANDON and HAZRA 1998), of which

563 species are endemic (CHANDRA and GUPTA, 2013). Whereas, Orthopteran fauna of Indian Mangroves consists of 48 species belonging to 35 genera under 8 families (CHANDRA and KUMAR, 2019). Orthopteran as a group, forms an important component of the food web, being eaten by amphibians, reptiles, birds, rodents, bats, primates, and insects and are also known to occur in a wide variety of habitats, ranging from the littoral zone of the seashore to grasslands, forests, and mountain tops, well above the tree line. The purpose of this study is to procure some basic information on Orthoptera fauna as no study was conducted previously, and the data will serve as baseline information for future research.

MATERIALS AND METHODS

Two surveys were conducted at Lothian Island (Fig-1) one in winter and the other in summer of 2019-20. After a preliminary survey, the Lothian Island was hypothetically divided into three zones for sampling of Orthoptera specimens, the northern zone, south and the south eastern zone. In the northern and southern parts most of the regions were covered with dense mangrove forest, and South-Eastern part had a different habitat covered with mudflats along with grassland and mangrove plants.

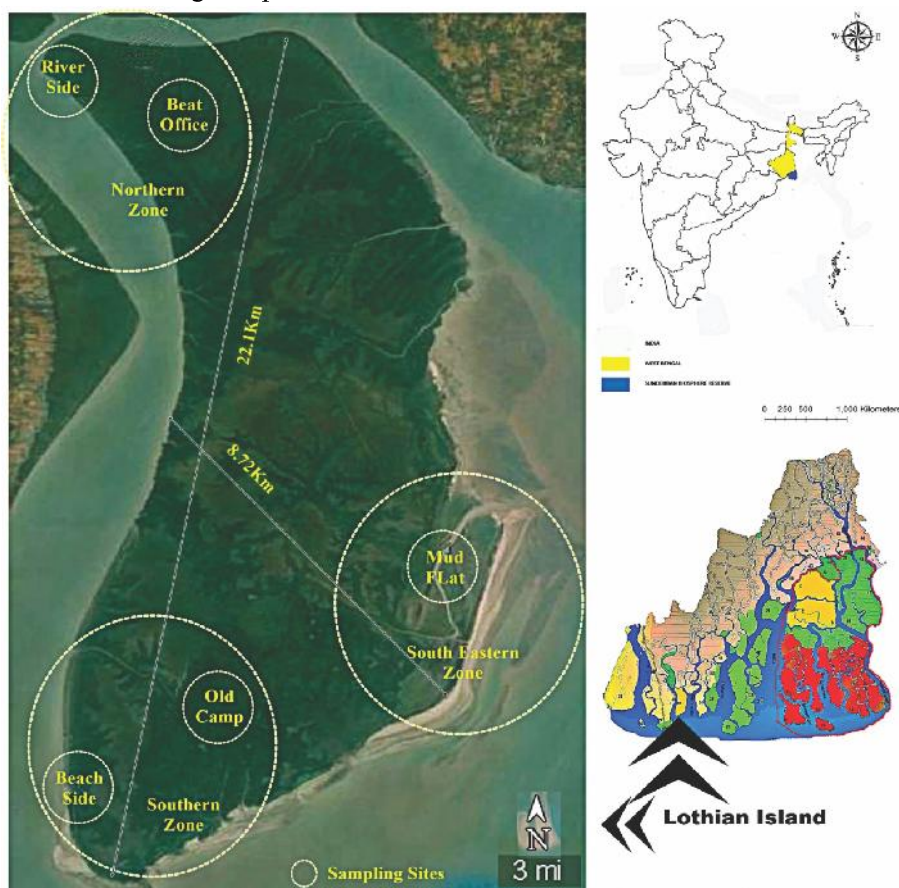


Fig-1 Survey localities in Lothian Island WLS, Sunderban Biosphere Reserve, W.B.

Orthopterans were collected from different localities of Lothian Island. Sweep net and search out methods were employed during daytime, and light trap was used for nocturnal

insects. During night collection, white cloth (2m x 2m) was hung from the tree top (near by 2 meter above ground level) with a 40 watt LED bulb connected to an Inverter (Intex, UPS-50, 160 -230 volts). The collected specimens were studied with the help of available literature under the Stereo Zoom Binocular Microscopes, model Leica EZ4 HD. For identification of species and distributional details, The Indian Checklist of SHISHODIA *et al.* (2010) has been followed. Identified specimens were registered and deposited in the National Zoological Collections of Zoological Survey of India, Kolkata & listed in Table. Relative abundance was estimated by the formula laid by PRESTON, 1948.

RESULTS AND DISCUSSION

A total of 151 specimens of Orthoptera were collected from three different zones, northern, southern, and south-eastern zones of Lothian Island (Fig-1). Thirty species of Orthopteran were identified from the samples collected belonging to 25 genera under 15 sub families and 6 families (Table). Family Acrididae and Tettigoniidae were the most represented with 10 and 7 species respectively (Fig-2), followed by Tetrigidae and Gryllidae with 4 each. Pyrgomorphidae was the least represented with only 2 species viz., *Atractomorpha crenulata* and *Tagasta indica*. The Orthopteran diversity in mangroves may vary from one region to another. MING KAI TAN, 2013 worked on the Orthopteran fauna in mangroves of Singapore and recorded six families of Orthoptera. Family Gryllidae represented maximum number of species 43%, while the other five families had less than 20% of species representation. However, in Sunderban Acrididae showed the maximum number of individual abundance 95 specimens belonging to 10 species. The three species which were predominantly more abundant were *Aiolopus thalassinus tumulus*, *Spathosternum pr. prasiniferum* and *Oxya fuscovittata* compared to that of any other species. Lothian Island represents only 30 species (10.8%) (Table) of the 278 Orthopteran species occurring in the state of West Bengal (CHANDRA, 2011), despite the small area there are three species which are new records to the state of West Bengal, *Ducetia dichotoma* Ingrisch and Shishodia, 1998, *Hexacentrus unicolor* Serville, 1831, and *Pusana laevis* (Uvarov, 1921), while five species are new to Sunderban Biosphere Reserve, *Conocephalus (Anisoptera) maculatus* (Le Guillou, 1841), *Tagasta indica* Boliver, 1905, *Xenocatantops humilis* (Serville, 1838), *Hebarditettix quadratus* (Hancock, 1915), and *Loxoblemmus equestris* Saussure, 1877.

Orthopteran collected from the three different zones of Lothian islands varied with respect to the abundance and species richness (Fig-3). Northern zone of the island showed more than 50% for both individual abundance as well as species richness. The south eastern zone was relatively less favorable spot for the Orthopteran as its representation for diversity and abundance was less than 10%. Considering the two major habitats grasslands and mangrove ecosystem, the Orthopteran were available in both these habitats, however its representation in the mangrove was more, 69% against 31% in the grasslands (Fig-4). The relative abundance as depicted in Fig-5 is relatively high in three species of the family Acrididae viz., *S. prasiniferum*, *A. thalassinus tumulus*, and *O. fuscovittata*.

Interestingly, these three Acridids are also rice feeding grasshoppers occurring in other islands of Sunderban (USMANI *et al.*, 2012; CHAND and NANDY, 2016; and DAS *et al.*, 2016). In Sunderban there are 104 islands out of which 54 are completely deforested and inhabited by humans, where rice is the most important food crop for the people of Sunderban. Rice plays a major role in the food as well as the livelihood security of the farming community of Sunderban.

Table: List of identified Orthoptera from Lothian island WLS

S. No	Species (Subfamily: Family)	Specimens identified
1	<i>Acrida exaltata</i> (Walker, 1859) (Acridinae: Acrididae)	1
2	<i>Phlaeoba infumata</i> Brunner Von Wattenyl, 1893 (Acridinae: Acrididae)	1
3	<i>Aiolopus thalassinus tumulus</i> (Fabricius, 1798) (Oedipodinae: Acrididae)	31
4	<i>Pusana laevis</i> (Uvarov, 1921) (Oedipodinae: Acrididae)	1
5	<i>Spathosternum prasiniferum prasiniferum</i> (Walker 1871) (Spathosterninae: Acrididae)	30
6	<i>Oxya nitidula</i> (Walker, 1870) (Oxyinae: Acrididae)	5
7	<i>Oxya fuscovittata</i> (Marschall, 1836) (Oxyinae: Acrididae)	23
8	<i>Oxya hyla</i> Serville, 1831 (Oxyinae: Acrididae)	1
9	<i>Xenocatantops humilis</i> (Serville, 1838) (Cantopsinae: Acrididae)	1
10	<i>Eyprepocnemis alacrisalacris</i> (Serville, 1839) (Eyprepocnemidinae: Acrididae)	1
11	<i>Tagasta indica</i> Boliver, 1905 (Pyrgomorphinae: Pyrgomorphidae)	1
12	<i>Atractomorpha crenulata</i> (Fabricius, 1793) 1905 (Pyrgomorphinae: Pyrgomorphidae)	7
13	<i>Coptotettix annandalei</i> Hancock, 1915 (Tetriginae: Tetrigidae)	8
14	<i>Ergatettix dorsifera</i> (Walker, 1871) (Tetriginae: Tetrigidae)	1
15	<i>Hedotettix gracilis</i> (de Haan, 1842) (Tetriginae: Tetrigidae)	2
16	<i>Hebarditettix quadratus</i> (Hancock, 1915) (Scelimeninae: Tetrigidae)	1
17	<i>Loxoblemmus equestris</i> (Saussure, 1877) (Gryllinae: Gryllidae)	3
18	<i>Modicogryllus (Modicogryllus) cofirmatus</i> (Walker 1859) (Gryllinae: Gryllidae)	5
19	<i>Teleogryllus (Macroteleogryllus) mitratus</i> (Burmeister, 1838) (Gryllinae: Gryllidae)	3
20	<i>Polionemobius taprobanensis</i> (Walker, 1859) (Nemobiinae: Trigonidiidae)	1
21	<i>Pteronemobius (Pteronemobius) heydeniicolor</i> (Walker 1871) (Nemobiinae: Trigonidiidae)	1
22	<i>Natula longipennis</i> (Serville, 1838) (Trigonidiinae: Trigonidiidae)	1
23	<i>Trigonidium humbertianum</i> (Saussure, 1878) (Trigonidiinae: Trigonidiidae)	3
24	<i>Conocephalus (Anisoptera) longipennis</i> (Haan, 1843) (Conocephalinae: Tettigonidae)	1
25	<i>Conocephalus (Anisoptera) maculatus</i> (Le Guillou, 1841) (Conocephalinae: Tettigonidae)	3
26	<i>Euconocephalus pallidus</i> (Redtenbacher, 1891) (Conocephalinae: Tettigonidae)	8
27	<i>Ducetia japonica</i> (Thunberg, 1815) (Phaneropterinae: Tettigonidae)	3
28	<i>Ducetia dichotoma</i> (Ingrisch and Shishodia, 1998) (Phaneropterinae: Tettigonidae)	2
29	<i>Holochlora</i> sp. (Phaneropterinae: Tettigonidae)	1
30	<i>Hexacentrus unicolor</i> Serville, 1831 (Hexacentrinae: Tettigonidae)	1

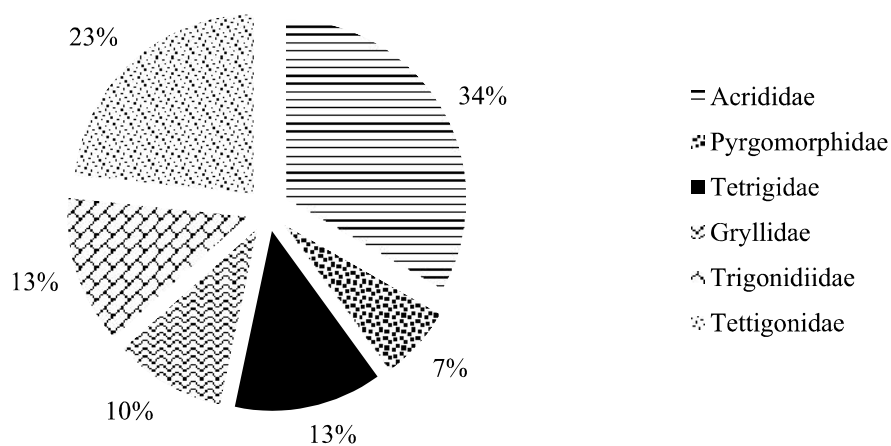


Fig- 2: Family wise Orthopteran spp. distribution in Lothian Island WLS, SBR, WB

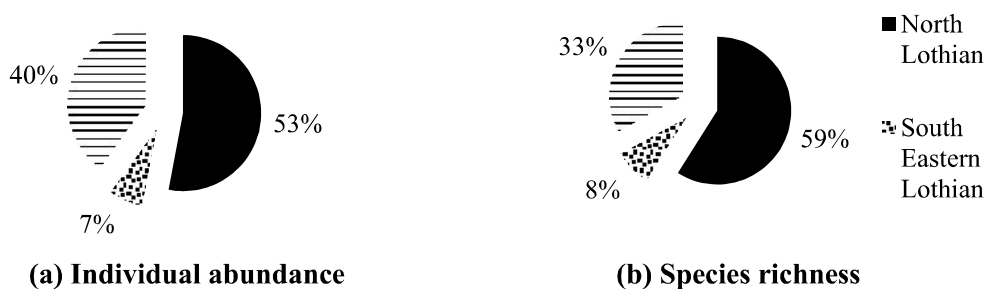


Fig-3: Distribution of Orthoptera in different zones of Lothian Island WLS, SBR, WB

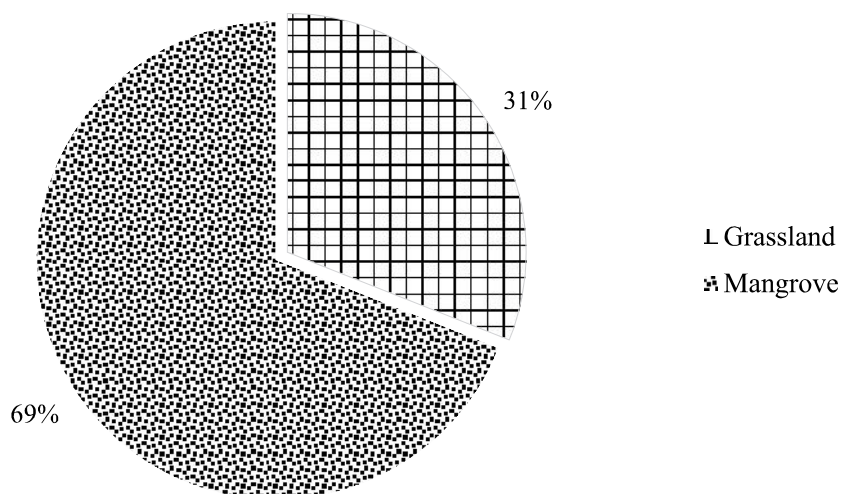


Fig-4: Habitat wise Orthopteran spp. distribution in Lothian Island WLS, SBR, WB

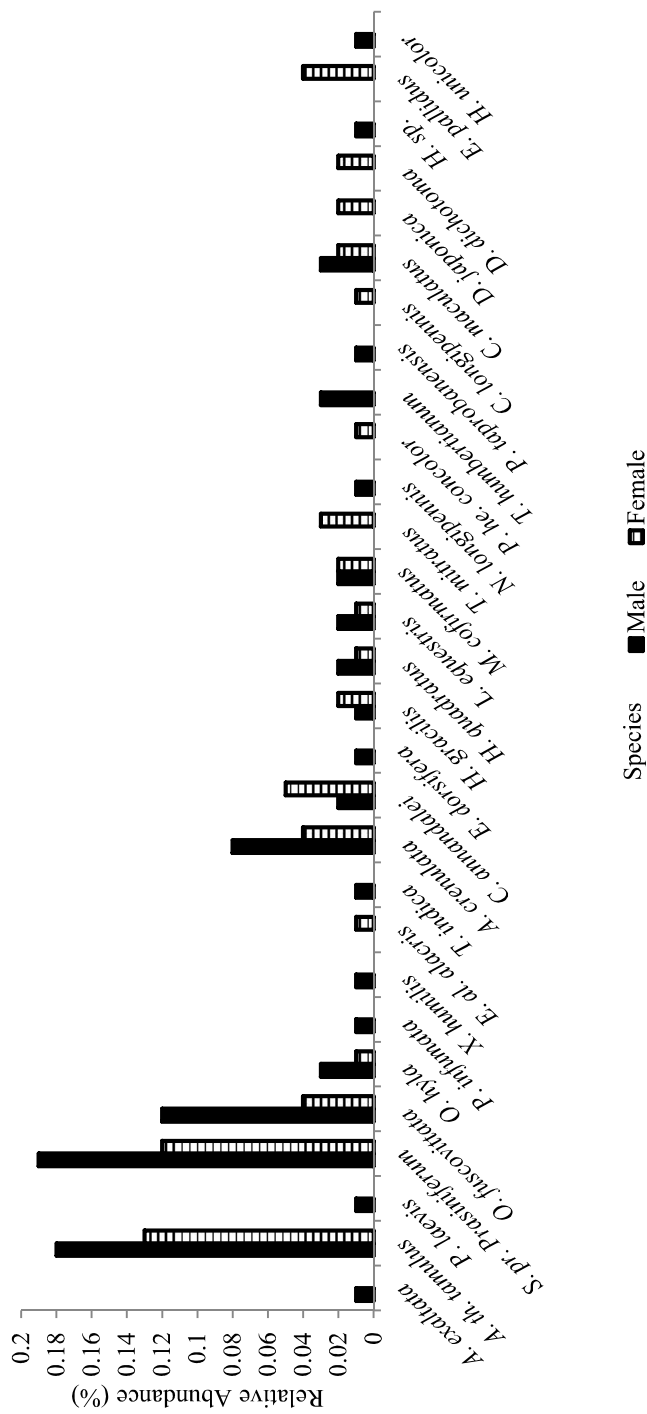


Fig-5: Relative abundance of male and female of Orthopteran

A. thalassinus tamulus was found to be present throughout the year in all parts of the agricultural eco-system of Sunderban (CHAND and NANDY, 2016). *Aiolopus thalassinus* is a long-winged grasshopper with strong flight capacity and it also shares some of the density-dependent behavioral and physiological characteristics of the migratory locusts (ELMQVIST *et al.*, 2020). More research on the aspects of bio-ecology and insect plant inter relationship in the mangrove would be helpful in understanding the overall diversity in the mangrove forests of Sunderban, and also enable researches to compare the Orthopteran fauna in mangrove patches and adjacent vegetation. Insects are also often used in environmental studies as bio-indicators due their diversity, abundance and close relationship with biotic and abiotic factors in environment. They can indicate the effects of habitat changes and fragmentation, and the effectiveness of management schemes designed to preserve or change individual species or community level patterns (CHUNG, 2013). Such studies will give an understanding of Sunderban mangrove forests & an alternate perspective of the conservation status and value of different mangrove patches.

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